

B.Tech. 4th Sem Electrical Engg – Control Systems EEA1-4 Second Sessional 07/04/2025

Max marks: 15

Time: 50 min

Attempt All Questions

1. Design a Lead Compensator for system $G(s) = \frac{K}{s(s+1)}$ so that unity feedback close loop system has $\zeta = 0.7$, settling time $t_s = 1.4$ secs. 5

2. Obtain the expression for the resonant peak and resonant frequency for a system 5

$$\text{with } G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

4

3. Given a system $G(s)H(s) = \frac{1}{s(1+2s)(1+s)}$. Sketch roughly the Nyquist plot & compute gain margin. Also comment whether system is stable in close loop or not? 6